

Qatar-1b

R-2789

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130609 · created 2026-09-04T00:48:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456452.786974 +/- 5.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.02
Transit depth (Rp/Rs)^2	3.07 +/- 0.69 [%]
Orbital Inclination (inc)	83.01 +/- 1.99
Airmass coefficient 1 (a1)	1.29 +/- 0.017
Airmass coefficient 2 (a2)	-0.19 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	None
Optimal Aperture	3.56
Optimal Annulus	18.55
Transit Duration (day)	0.061 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.02 vs 0.1463 (deviation 1.43σ)
Duration fit vs published	0.061 d vs 0.0692 d (deviation 0.45σ)
Mid-time O–C	0.1 min
Depth SNR	8.8
Residual scatter	1.28 %

Light curve

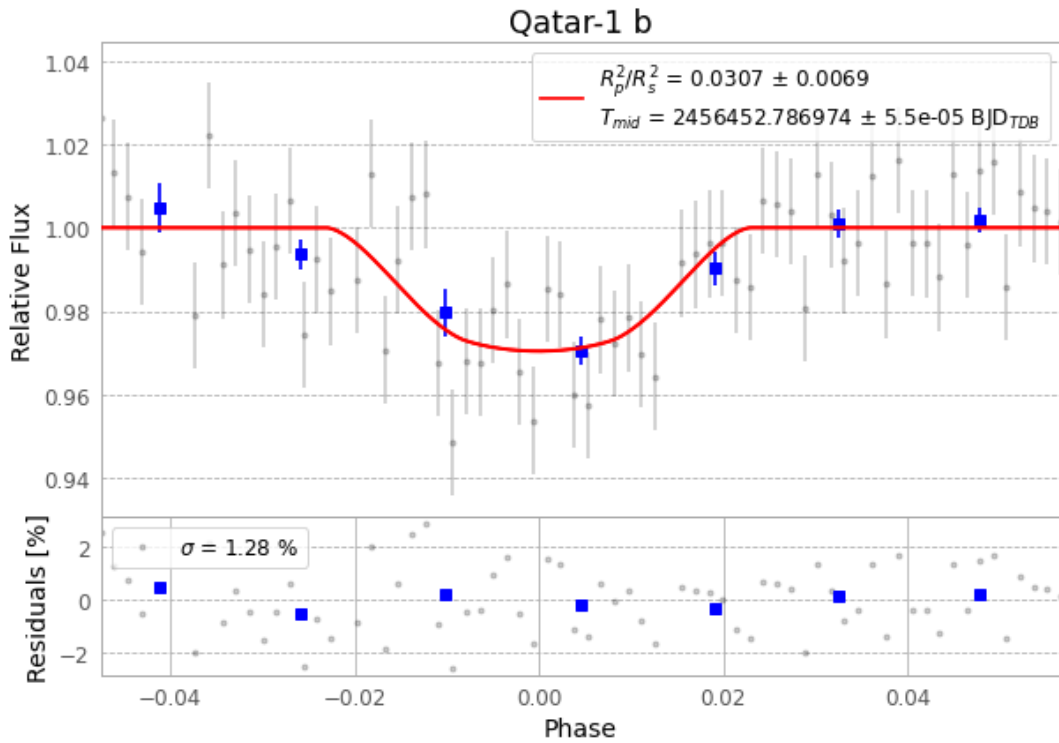


Figure 1 — FinalLightCurve_Qatar-1 b_2013-06-08.png (SHA-256 24bbffbf033e991...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [422, 169], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 453efaac31693560...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1130609051512.FITS → Qatar-1130609084812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130609.log (f8265123cd18...), FinalLightCurve_Qatar-1 b_2013-06-08.png (24bbffbf033...), FinalLightCurve_Qatar-1 b_2013-06-08.csv (cf07e16bc449...)

Compute resources

Wall time 172.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 00:48Z.